Quiz 5

Friday, October 9 — basis, dimension, and orthonormal vectors

Last updated: October 5, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Related lecture: Basis (Week 5). Same day: Transforms and Lab 3 (not on this quiz).

Textbooks: Boyd & Vandenberghe (VMLS) · also on Resources.

Schedule

Core

- B 5.3 (basis expansion / coefficients in a given basis)
- B 5.4 (norm of a linear combination of orthonormal vectors)
- C5.1: Decide whether a short list is a basis of $\mathbb{R}^2$ or of a named subspace; if not, say whether it fails independence, spanning, or both.
- C5.2: Find a basis and the dimension of a span given by three vectors with an obvious redundancy.
- C5.3: Check whether a pair is orthonormal; if it is, find the orthonormal coefficients of a given v.

Stretch

B 5.5; B 5.6 (Gram–Schmidt on a structured list — light).

What a quiz item might change

Which list is proposed as a basis, one redundant generator in a span, or a different orthonormal pair and target vector.

Note

Matrices, Ax, and transformations (Lectures 18–20 / Lab 3) start this week and are **not** on this quiz; they are on Quiz 6.